AMRITA VISHWA VIDYAPEETHAM
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Type 1: Research Project

Evolutionary Meta-Optimisation of Deep Reinforcement Learning for Cryptocurrency Trading

Team Number: [100]

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Literature Survey (1/4)

- **Survey Scope:** Analyzed key papers in Evolutionary Computation, DRL, and Financial Trading.

Paper Name	Key Findings	Problem Addressed
Comparative Study of PPO and DQN [1]	RL directly optimizes sequential actions; PPO captures momentum aggressively while DQN controls risk.	Lack of direct algorithmic comparisons constrained by strict market trend filters.
Financial Trading Enhanced by Cluster Embedding [2]	Adding predicted future trends and technical indicators into the RL state space drastically improves trend capture.	Existing models rely too heavily on lagging historical data, failing during sudden regime shifts.
Practical Approach to Address Backtest Overfitting [3]	Combinatorial cross-validation is critical to mathematically prove the agent survives actual crypto crashes.	Crypto RL research uses simple cross-validation methods that suffer from data leakage and false-positive reporting.
Optimization of Crypto Trading Strategies [4]	Dynamic reward functions penalizing severe drawdowns force the agent to learn strict risk-control behaviors.	Traditional static forecasting models suffer from a "decision disconnect" in dynamic execution.

Literature Survey (2/4)

Paper Name	Key Findings	Gap Identified
Efficient Evolution of Neural Network Topologies [5]	Introduced NEAT; evolves weights and topology; uses historical markings for crossover; speciation protects innovation.	Earlier methods failed to combine structures; innovations removed prematurely; random topologies wasted search time.
Evolution-Guided Policy Gradient in RL [6]	Introduced ERL; combines EA + DRL; EA improves exploration; DRL learns via gradient descent from shared data.	DRL struggles with sparse rewards and brittle convergence; EAs are slow with high sample complexity.
Hybrids of RL and EC in Finance: A Survey [7]	Provides taxonomy for EC+RL hybrids; EC handles non-differentiable metrics; RL handles sequential decisions.	No comprehensive survey bridging RL, EC, and finance; models use unrealistic assumptions.

Literature Survey (3/4)

Paper Name	Key Findings	Gap Identified
Population Based Training of Neural Networks [8]	Introduced PBT; optimizes weights + hyperparameters jointly; replaces weak models with mutated top models.	Sequential tuning is slow; grid search wastes computation; fixed hyperparameters are suboptimal.
Differentiable Evolutionary Reinforcement Learning (DERL) [9]	Introduces gradient-informed mutations; uses differentiable signals to guide mutations; combines evolutionary exploration with gradient optimization.	Traditional EAs rely on random mutations (high complexity); RL alone struggles in non-stationary markets; lack of guided hybrid approaches.
TensorNEAT: A GPU-Accelerated Library for NEAT [10]	Moves evolutionary computation to GPU via tensorization; enables parallel evaluation of thousands of agents; uses JIT compilation.	EAs are computationally expensive; CPU-based evaluation limits its scalability; lack of efficient frameworks for real-time high-frequency evolution.

Literature Survey (4/4)

Paper Name	Key Findings	Gap Identified
Transformer-based ERL for Multi-Regime Trading [11]	Integrates attention-based learners with evolutionary layers to detect market shifts; proves hybrid stability across diverse regimes.	Standard RL agents suffer from "catastrophic forgetting" during regime shifts; lack of architectural memory for Bull/Bear transitions.
Topological Transfer Learning for Cross-Asset Trading [12]	Demonstrates that evolved neural topologies (brain shapes) are more transferable and robust across assets than fixed weights.	Weight-only transfer learning fails in high-volatility assets; fixed architectures cannot generalize across varying asset classes (e.g., BTC to ETH).

1. Literature Survey (1/3)

Rubric: Survey Extensiveness & Justification

- **Key Findings for RL:**
  - *Action Optimization [1]:* RL directly optimizes sequential Buy/Hold/Sell actions for BTC/ETH returns rather than merely predicting future prices.
  - *Mitigating Overfitting [3]:* Combinatorial cross-validation is mathematically required to prove the agent survives actual crypto crashes.
  - *State-Space Augmentation [2]:* Enhancing raw OHLCV with technical indicators drastically improves adaptation to sudden market regime shifts.
  - *Risk-Aware Rewards [4]:* Dynamic reward functions that explicitly penalize severe drawdowns enforce strict risk-control behaviors.

- **Algorithm Selection (Why PPO?):** We selected PPO over DQN because PPO natively supports stable policy updates and captures aggressive bullish momentum much more effectively in highly volatile crypto markets [1].

1. Literature Survey (2/3)

Key Findings in Neuroevolution and RL

- **Key Findings:**
  - *NEAT [5]:* Evolves network structures incrementally for hidden states; lacks gradient-based sample efficiency.
  - *ERL [6]:* Population approach overcomes reward sparsity; assumes fixed hyperparameters.
  - *Finance Hybrids [7]:* Combines EC with RL for financial metrics; notes failures due to unrealistic assumptions.
  - *PBT [8]:* Discovers dynamic hyperparameter schedules on-the-fly; prevents convergence failures.
  - *TensorNEAT [10]:* Leverages GPU tensorization to overcome traditional CPU scalability bottlenecks.
  - *DERL [9]:* Replaces random search with gradient-informed mutations for higher sample efficiency.

1. Literature Survey (3/3)

Addressing Market Volatility and Scalability

- **Research Gaps:**
  - **Static Fragility:** Reliance on fixed architectures and hyperparameters that fail in non-stationary markets [3].
  - **Scalability Bottlenecks:** Significant delays in high-frequency data processing on traditional CPU-based neuroevolution.
  - **Efficiency Constraints:** Heavy reliance on random mutations leads to poor sample efficiency compared to modern PPO approaches [1].

- **Proposed Direction:**
  - We aim to solve these limitations using a unified **TensorNEAT-PPO framework** with **Differentiable Evolutionary logic**.
  - This methodology dynamically adapts neural architectures and policies on-the-fly, enabling high-throughput, sample-efficient trading in volatile cryptocurrency environments.

2. Originality of Work

Novel Hybrid NE-PPO Framework

Originality Statement

Introduces a **Neuro-Evolutionary PPO** framework—the first to use **NEAT** structural evolution to optimize **PPO** topologies for non-stationary crypto market.

I. Automated Topology (NEAT+PPO)

- *Innovation:* Replaces fixed PPO layers with augmenting topologies.
- *Gap:* Solves "Architecture Sensitivity"; prevents overfitting in noisy data.

II. Diversity-Enhanced Exploration

- *Innovation:* Uses population-based search alongside PPO gradient descent.
- *Gap:* Escapes "Local Optima Traps" by maintaining a library of regime-specific strategies.

III. Tensorized GPU Hybridization

- *Innovation:* Vectorized PPO objectives within **TensorNEAT**.
- *Gap:* Shifts CPU bottlenecks to GPU; enables 1,000+ agent evolution in minutes.

IV. Gradient-Informed Structural Evolution

- *Innovation:* Directs NEAT mutations using PPO gradient signals for non-stochastic architecture growth.
- *Gap:* Overcomes "Black-Box" search inefficiency by targeting structural bottlenecks via local RL feedback.

3. Technical Knowledge (Algorithms/Models)

Rubric: Demonstrated Knowledge of Fundamentals

Proposed Mathematical Model / Algorithm:

- **Core Algorithm:** Proximal Policy Optimization (PPO) for Trade Execution.
- **Justification:**
  - **Clipped Surrogate Objective:** Prevents destructively large policy updates, ensuring stable learning and preventing the agent from overfitting to sudden crypto market noise.

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]$$

  - **Momentum Capture:** As an on-policy actor-critic algorithm, it adapts rapidly to recent market dynamics, aggressively capturing bullish momentum and trends [1].
  - **Continuous Action Spaces:** Can be configured to output precise portfolio allocation percentages rather than just binary Buy/Sell signals.

PPO Algorithmic Flow for Crypto Trading

Observe Market State $\rightarrow$ Actor Outputs Action $\rightarrow$ Critic Evaluates Value $\rightarrow$ Compute Advantage $\rightarrow$ Clipped Policy Update $\rightarrow$ Loop

3. Technical Knowledge (Algorithms/Models)

Rubric: Demonstrated Knowledge of Fundamentals

Proposed Mathematical Model / Algorithm:

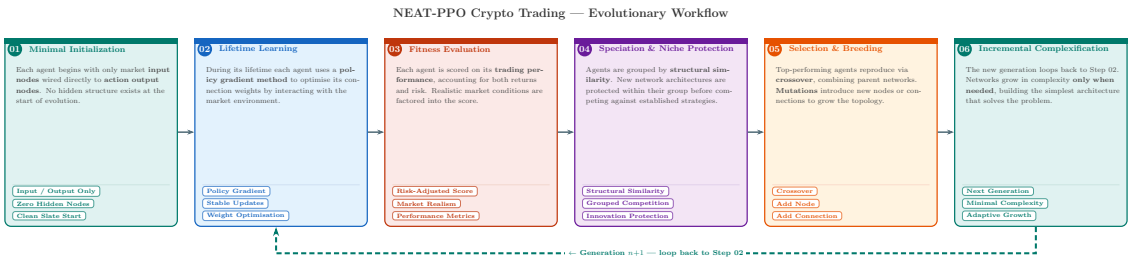
- **Core Algorithm:** NeuroEvolution of Augmenting Topologies (NEAT) for DRL.
- **Justification:**
  - **Complexification:** Starts minimal, adding nodes/links only when needed to handle market volatility (prevents overfitting).
  - **Speciation:** Protects novel, experimental trading strategies from early elimination before their weights optimize.
  - **Historical Markings:** Ensures stable crossover of different network structures by aligning innovation numbers.

NEAT Algorithmic Flow for Crypto Trading

Init Minimal Networks $\rightarrow$ Evaluate Fitness (Sharpe Ratio) $\rightarrow$ Speciate $\rightarrow$ Breed & Mutate $\rightarrow$ Loop

4. Proposed System Workflow

NEAT-PPO Evolutionary Loop — Left to Right



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Thank You!

Questions?